ARC-AGI-2 × Qwen3.8-27B

Full Project Handoff / Research Memory

গুরুত্বপূর্ণ findings, corrections, failed assumptions, validated results, source provenance এবং next-step plan

Snapshot: 2026-09-16 | Prepared for a new chat

Current state: Engineering/research pipeline আনুমানিক 60–65% complete; competitive solver readiness আনুমানিক 40–45%. Systems, provenance, reasoning-control এবং serving bottlenecks এখন অনেক পরিষ্কার; score-producing verified solving, multi-sample search ও Public-120 scaling এখনও সামনে।

How to use this handoff

এই PDF নতুন chat-এর canonical context হিসেবে ব্যবহার করুন। “VERIFIED” = executed artifact/source audit দ্বারা সমর্থিত; “INFERENCE” = evidence-supported interpretation; “RETIRED” = আগের claim/plan পরে ভুল/দুর্বল প্রমাণিত। নতুন chat যেন rejected notebook বা superseded assumption থেকে আবার শুরু না করে।

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1. Objective, environment, frozen controls

লক্ষ্য: Kaggle ARC-AGI-2-এ Qwen3.8-27B-FP8 দিয়ে reproducible agentic/program-synthesis solver তৈরি করা, যেখানে reasoning control, Python execution, deterministic train verification, candidate diversity এবং serving throughput একসাথে optimize হবে।

Item	Canonical value / rule
Model	Qwen3.8-27B-FP8, offline Kaggle checkpoint
Hardware	Kaggle 4× NVIDIA L4
Serving baseline	vLLM 0.19.0, TP=4
Pinned stack	Transformers 4.57.6; tokenizers 0.22.2; HF Hub 0.36.2
Context	29,696 max_model_len in Q1 engine
Q1 engine	enforce_eager=True; custom all-reduce off; prefix caching off
Frozen Proxy-48 SHA256	8f432c60a35091dcda2b890574b1f4d3c03aaf48a5b3b7f96b4e88c35d359134
Scientific discipline	একবারে এক mechanism; same frozen tasks + matched budget/time where causal comparison matters
Correctness authority	Official exact-grid + fresh all-train exact executable verification; raw cell similarity diagnostic only

Global implementation order from the audited Master Plan/portability notes: Direct baseline → PIV stateful REPL → PTQ fresh-call deterministic repair → code-family decision; LANG/CORE/LIB/Judge/heavier multi-

agent/evolution পরে। Empirical results পরে reasoning-control ও throughput sub-experiments যোগ করেছে, কিন্তু এই causal discipline অপরিবর্তিত।

2. Progress snapshot

Stage	Status	Meaning
Baseline / forensic	DONE	Q1 OFF/XHIGH failure signatures, exact scoring, provenance established.
Executor / verifier	DONE / hardened	Persistent REPL, sandbox, retry, train verifier, budget/checkpoint/provenance repeatedly tested.
Reasoning→action	MOSTLY DONE	LOW and hard thinking-budget enter tool phase; MEDIUM weak. Solve quality still pending.
Serving optimization	ACTIVE, strong signal	CUDA graph/noAR major gain; AR negligible; prefix/TP2 need focused retest.
Verified solving	CRITICAL PENDING	Need consistent final programs, all-train-exact >0 and official exact solves.
Multi-sample / pass@K	PENDING	Likely major coverage lever once single-lineage solver is healthy.
Public evaluation	PENDING	Proxy/Public-48 then Public-120 after solver-smoke success.
Submission scaling	PENDING	Only after reproducible score/runtime evidence.

Progress estimate: Work volume remaining ~35–40%, but it is the score-producing part: solver integration, candidate search, verified accuracy and public evaluation.

3. Q1 Direct baseline — diagnostic ground truth

Q1 Direct OFF vs XHIGH on frozen Proxy-48 was engineering-clean but solver-failing. It is the strongest baseline for understanding the model's failure mode.

Metric	OFF	XHIGH
Task generations	48/48	48/48
Test outputs	56	56
Official exact	0/56	0/56
Task parse-valid	29/48 = 60.42%	0/48
Valid output grids	31/56 = 55.36%	0/56
Length-cap	18/48 = 37.5%	48/48 = 100%
Generated tokens	106,438	589,824
Generation behavior	Some natural stops / usable grids	12,288-token reasoning to cap; no usable

		final

Total runtime 14,007 s ≈ 3 h 53 m. Independent saved prediction vs ground-truth exact-array comparisonও 0 exact confirm করেছে; scoring bug ছিল না। XHIGH মোট generated tokens-এর ~84.7% এবং generation time-এর ~84.8% খরচ করেছে, usable candidate 0।

Durable Q1 lessons

- Direct XHIGH = termination/commitment failure: 48/48 length-cap. Blindly raising 12k→16k was not evidence-based priority.
- OFF natural-stop parsing largely worked; overall parse loss mainly length-cap, not parser architecture.
- Dynamic / Tier-3 / high-ratio / extreme-ratio / multi-query slices disproportionately collapsed, pointing toward external computation/verification need.
- Raw cell-match can be inflated by unchanged background/input structure. Use exact, changed-cell diagnostics, and executable train verification.
- A manually reconstructed “near-solve” from reasoning prose was later RETIRED as an experimental metric because it was not a saved final candidate.

4. PIV lineage — harness hardening and XHIGH activation failure

PIV-Q27-P0 intended loop: one lineage → persistent Python REPL → observe → revise → self-contained solution(grid) → fresh subprocess every-train exact → test execution. This was chosen because it directly targets Q1’s unverified monologue failure.

PiVOTools source provenance

Artifact	Verified source
Repo commit	gutfeeling/arc-agi-2-submission @ b3d4494892c6d85fa2a35d548271fb02ea3e191b
Agentic prompt blob	dab87e08bbe3f071920f13eb98f4a4d6f03ada10
System prompt blob	265131a50fb42deebe0a8da353f466b11d6d7d6b
Retry semantics	Inner same-history retry for malformed/incomplete; outer turn retry after exhaustion
Final artifact	CodeSolutionTurn requires solution; solver verifies all training examples exactly before test
Important mapping rule	Upstream GPT-OSS “high” is not a Qwen effort value; string names must not be mapped naively.

Key PIV hardening lessons

- v1.4 corrected retry hierarchy and verifier sentinel parsing.
- v1.5 PREFLIGHT later exposed malformed-tool retry reset, missing-terminal coverage, stale metadata.
- v1.6 added stratified SMOKE8, guard accounting, tool-output sanitization, real scheduler fake preflight, and local final-on-length salvage.

- v1.7 fixed a real multi-call scheduler bug: guard-rejected calls must not consume the remaining valid execution slot; admission became dynamic on executed tool_calls.
- Length-final salvage is LOCAL adaptation, not upstream PiVOTools behavior; must be logged separately and accepted only through fresh exact verification.

v1.7 real SMOKE_GPU

Metric	Result
Tasks	8/8
Infrastructure failures	0
First-turn length	8/8 = 100%
Generated tokens/task	12,288/12,288
Tool engagement	0/8
Final program	0/8
All-train exact	0/8
Budget-death	100%
Generation wave	1,967.8 s ≈32.8 min
Projected Proxy-48	~3.40 h; +20% ~4.06 h

Correct conclusion: “PIV কাজ করে না” নয়। VERIFIED: Qwen3.8 XHIGH + current template + 12,288 generated-token budget-এ model tool phase-এ পৌঁছায়নি। Throughput তখন primary blocker ছিল না.

5. Reasoning activation ablation

Major correction: Qwen3.8 accepted effort values are xhigh / medium / low. Earlier “HIGH smoke” recommendation was RETIRED. Qwen hard thinking-budget uses documented two-stage continuation: budget পর্যন্ত reasoning; natural close না হলে early-stop sentence + </think>; then exact token-prefix continuation.

Arm	Intervention	Real GPU activation result
ACT-LOW	reasoning_effort="low"	Tool engagement 0.875; first tool ~21 think tokens; later turns sometimes 3,072 cut.
ACT-MEDIUM	reasoning_effort="medium"	Tool engagement 0.125; weak; rejected.
ACT-BUDGET3072	xhigh + 3,072 hard think budget + forced close	Tool engagement 0.875; 7/8 phase-2 substantive code.

Probe run 20260915T124541Z (~59 min). Approved arms: LOW and BUDGET3072. This proves reasoning→action activation, not puzzle-solving quality. Forced close must be analyzed by post-close action type; it does not guarantee a tool call.

Activation implementation rules that must persist

- Historical reasoning_content must not contain a duplicate literal </think>; replay template closes it.
- Injected stop tokens are not model-generated tokens, but they do consume context/prefill and must be logged.
- Phase-2 should use token-exact prefix continuation; no silent text re-render fallback.
- Harness/PREFLIGHT correctness and real-model behavioral activation must always be reported separately.
- Tool engagement alone is not success: Activation → Productivity → Executable Quality → Compute/Stability.

6. Throughput / serving optimization — latest state

Leaderboard screenshots showed notable score increases after Qwen3.8 release for several teams, but this is temporal correlation only; it does not prove their model or serving recipe. Our throughput work therefore measures our hardware directly rather than inferring competitors' implementation.

Probe design corrections before the latest run

- Long-decode anchor added to avoid apples-to-oranges comparison with short prefill-heavy waves.
- Custom all-reduce branched independently; batch/cache rows no longer silently inherit AR.
- TP4, TP2 single, synchronized TP2×2 and GPU pairings added.
- Prefix exact-repeat supplemented by realistic PIV/Qwen tool-history extension and next-agent-turn workload.
- Cleanup scoped to processes created by current config; evidence ZIP includes manifest, raw logs, telemetry, contract and handoff file.
- No raw-throughput row is automatically promoted to solver; real solver smoke remains required.

Latest completed real run: v1.2.4 SMOKE_GPU

Run ID 20260916T091344Z, 4×L4. Evidence ZIP SHA256: c34fd3736dd9d2565ef681c6fd1936bb7eb1f99b3a95d222eda419049cd3e2aa. 78 ZIP members; 77 evidence files + MANIFEST; 77/77 listed size/SHA verified. No Jupyter error output. Pinned inference stack later passed parity despite unrelated Kaggle package resolver warnings.

Config	decode@8	decode@16	mixed@16	long@8
Frozen Q1	27.15	37.68	23.59	48.30
Stats-control eager	24.56	37.73	23.71	—
CUDA graph, no AR	42.14	48.95	31.45	129.17
CUDA graph + custom AR	42.14	48.93	31.44	129.04

Strongest serving result: C1 graph/noAR long@8 = 129.17 tok/s vs frozen Q1 48.30 tok/s ≈ 2.67×. n=16 fixed ~+29.8%; mixed ~+32.7%. Custom all-reduce ≈0 gain and slightly higher resource use; keep AR off for now.

Failed rows — what they do and do not prove

- C5 prefix64 loaded and warmed successfully, then OOM at decode@32. Prefix caching itself was not disproven; cache measurement was confounded with max_num_seqs=64 / capacity stress.
- T1 TP2 noAR loaded and warmed, then OOM at decode@16. Logged KV cache 17,248 tokens, estimated max concurrency ~2.17× at 29,696. TP2 model fit is not disproven; chosen concurrency/prefill pressure failed.
- T2 TP2 AR had the same class of failure. Topology is not disproven; safe concurrency must be searched n=2→4→8 with smaller scheduler token budgets.

Critical v1.2.4 harness/analysis bugs discovered after the run

- Fail-fast bug: C5 and T1 wrote terminal failed/EngineDeadError early, but parent waited for child exit until full 75-min timeout. C5 4500.6 s + T1 4500.8 s + T2 1504.5 s = ~2.92 h, ~68% of the 4.29 h run wasted on failed rows. Next harness must watch partial result JSON and kill/reclaim shortly after terminal failure.
- Numerics gate invalid: unchanged end-of-run C0 repeat had greedy prefix agreement only 0.1875, below the 0.50 threshold; C1=0.1836, C2=0.4258. Therefore prefix agreement cannot be a promotion safety gate in this environment.
- Because the broken gate filtered C1, notebook's "C0 winner" was misleading. C1 graph/noAR remains the prime solver-smoke candidate.
- Memory-capacity ranking must not be numerics-filtered. Actual KV tokens: C0=159,936; C1=166,992; C2=167,776.
- Global scientific_comparison_valid=true was too broad. Family-level validity should be used: tp4_core_valid=true; prefix_valid=false; tp2_valid=false; tp2x2_valid=false.

Current throughput decision: Do NOT run the old broad PROBE_GPU profile. First validate C1 graph/noAR in a real solver smoke, and separately build a focused fail-fast throughput follow-up for prefix-at-mns16 and safe TP2 concurrency.

7. P1-SPARSE side branch (secondary, keep separate from main ARC-AGI-2 solver line)

A separate P1-SPARSE run was integrity-clean but not promotion-ready. Mean score 6.8091 vs historical B0 6.4940 (raw +0.3151); completed levels 39 vs 32 (+7); 17/25 games positive, 7 negative, 1 tie. Effect was high-variance, with large gains and large regressions.

- 24 read-timeout events and 77 vLLM preemptions were observed; all 25 games consumed near full per-game budget and ended "gave_up".
- Teardown left one vLLM GPU worker alive (~95 GB reported); artifacts survived, so run not invalid, but repeated-session cleanup is unhealthy.
- Comparison used a historical unmatched B0 reference. Promotion requires matched B0 vs P1 replication (preferably ≥2 matched pairs/seeds).
- Classify P1 as PROMISING / HOLD, not promoted and not rejected. Do not mix this branch's score logic with the main Qwen ARC-AGI-2 solver decisions.

8. Mistakes, invalid assumptions, and corrections

Mistake / weak assumption	Correction that should carry forward
"PiVOTools high → Qwen high"	Wrong. Qwen supports xhigh/medium/low; GPT-OSS high is a different scale.
PREFLIGHT pass implied model would use tools	Wrong. Harness correctness ≠ model behavioral activation; add tiny real mechanism-activation probes early.
Focus on parser/retry before first-tool behavior	Tunnel vision. First-turn 12,288 cap was the dominant signal; activation should have been measured earlier.
My first v1.8 notebook was safe because it compiled	Wrong. It had a runtime PROFILE_TABLE naming bug; compile/nbformat are insufficient without sequential execution/static name checks.
External v1.8 recommendation logic trusted insertion order	Bug: passing candidate had to be chosen from ranked candidates, not first arm order.
Force-close = guaranteed tool call	Wrong. It only forces exit from thinking; post-close can still be prose/malformed/final.
Throughput v1.0 "anchor" used short raw generate wall time	Not comparable to Q1 long behavior; long anchor needed.
Prefix/TP2 OOM = feature unsupported	Wrong. Aggressive concurrency/capacity stress confounded feature feasibility.
Greedy prefix agreement ≥0.50 = numerics safety	Invalidated by unchanged C0 repeat scoring 0.1875. Use real solver/all-train exact quality gates.
Global comparison_valid boolean	Too broad; family-level validity is required.
Wait for worker timeout after terminal failure	Wasteful; parent must detect terminal result JSON and fail-fast kill/reclaim.

9. What is proven vs still unknown

VERIFIED / high-confidence

- Q1 Direct exact coverage is 0; XHIGH direct is a catastrophic termination pattern under 12,288 tokens.
- PIV v1.7 XHIGH did not activate tools in 8/8 SMOKE tasks; this is model/template/budget interaction, not parser failure.
- ACT-LOW and ACT-BUDGET3072 can activate real tool use at 7/8 scale; MEDIUM is weak.
- CUDA graph (enforce_eager=False) materially improves 4×L4 serving; C1 graph/noAR is ~2.67× faster on long@8 than frozen Q1 in the latest smoke.
- Custom all-reduce gives essentially no speed gain on this L4 PCIe setup in tested conditions.
- Current prefix64 and TP2 OOMs are inconclusive about feature feasibility because concurrency pressure was too aggressive.
- Greedy token-prefix fingerprint is not a valid safety gate in this environment; unchanged control fails it.

UNKNOWN / requires experiment

- Does C1 graph/noAR preserve actual ARC program quality / all-train exact behavior?
- Which activation arm (LOW vs BUDGET3072) yields better executable program quality, not merely tool engagement?
- Can prefix caching at max_num_seqs=16 produce useful multi-turn speedup without OOM?

- Can TP2 operate at safe n=2/4/8 and smaller max_num_batched_tokens; if so, does TP2×2 beat TP4 aggregate throughput?
- What is the true pass@K / candidate diversity curve once a single-lineage solver produces verified programs?
- What Public-48 / Public-120 exact score is achievable within Kaggle time limits?
- Whether leaderboard jumps were caused by Qwen3.8 or a specific serving/agent recipe; screenshots alone cannot establish this.

10. Exact next-work sequence

Priority update: Latest v1.2.4 audit supersedes “run full throughput PROBE next”. Do not run the old broad PROBE_GPU.

1. C1 graph/noAR REAL SOLVER SMOKE. Change only engine serving from frozen eager to enforce_eager=False; custom AR off; prefix off. Keep a known reasoning-control arm and the same executor/verifier. Goal: prove speedup does not damage train-exact/solver behavior.
2. Focused throughput follow-up (new fail-fast harness). Isolate prefix caching at max_num_seqs=16; run cache A/B before capacity stress; TP2 n=2→4→8 with 8k/16k scheduler token budgets; only build TP2×2 from a safe TP2 point. Parent watches terminal result JSON and kills/reclaims immediately after failed/unsupported status.
3. Reasoning-control solver comparison. On the same small frozen task set, compare ACT-LOW vs ACT-BUDGET3072 for Activation → final-program rate → all-train-exact → official exact, with matched generated-token/time accounting. MEDIUM stays rejected unless new evidence.
4. Only after a real solve smoke shows non-zero verified programs, run Proxy/Public-48. Use the winning serving/control recipe; no broad architecture sweep inside the scoring run.
5. Add multi-sample candidate generation (K>1), fresh train verification, deduplication and pass@K/oracle coverage. Track unique task IDs and compute per unique rescue.
6. Run Public-120 when Proxy/Public-48 shows reproducible exact solves and runtime projection is acceptable. Then plan hidden/submission scaling.
7. PTQ-Q27-T0 remains the paired alternative if PIV/control line cannot deliver verified programs efficiently. Compare PIV vs PTQ on same frozen subset and matched budget before selecting the main code strategy.

Suggested gates for the next solver smoke

Layer	Gate / metric
Integrity	100% terminal coverage; 0 infrastructure corruption; bundle manifest/hash clean
Activation	Real Python/tool engagement and no first-turn runaway on most tasks
Productivity	Final-program rate >0; syntax-valid solution(grid) artifacts
Quality	all-train-exact >0 is the first strong milestone; official exact solves are the ultimate metric
Serving safety	C1 vs C0 matched solver outputs/training behavior; no quality regression hidden behind throughput

Compute	Measured wall time + projection with margin; no failed-row timeout waste
Diversity	Code-hash/prediction diversity, duplicate ratio, per-task candidate count
Slices	Easy/static, dynamic, multi-query, tier/ratio segments kept separate

11. Do-not-do list

- Do not run current/old broad PROBE_GPU before focused fail-fast redesign.
- Do not run PIV v1.7 XHIGH Proxy-48; its smoke gate correctly rejected it.
- Do not reintroduce Qwen reasoning_effort="high".
- Do not treat tool engagement as solving success.
- Do not use raw cell similarity or 4×64 greedy prefix agreement as correctness proof.
- Do not conclude prefix caching or TP2 are unsupported from the aggressive OOM rows.
- Do not turn custom all-reduce on by default; measured benefit ≈0.
- Do not silently enable prefix caching in scientific comparisons unless it is the explicit intervention.
- Do not change multiple major mechanisms in one causal experiment unless it is an explicitly pre-registered factorial comparison.
- Do not let fake/selftest success substitute for real GPU/model behavior.
- Do not wait full config timeout after terminal worker failure; fail-fast and reclaim.
- Do not infer competitors' exact model/throughput/recipe from leaderboard timing alone.

12. Canonical sources and provenance

ID	Source / artifact	Why it matters
S1	ARC_AGI2_Qwen38_Experiment_Master_Plan_2026-09-13	Global experiment order, one-mechanism discipline, PIV/PTQ matched comparison, metrics.
S2	ARC_AGI2_Agentic_Solver_Portability_Notes_v10_Final_English_2026-09-13	Source-audited portable solver mechanics; PIV implementation/verifier rules.
S3	Q1-DIR-Q27-PROXY48-v1.2-PRODUCTION.ipynb + Q1 bundle	Canonical Direct baseline and frozen Proxy-48.
S4	Frozen Proxy-48 SHA256 8f432c60…59134	Task-order integrity.
S5	PiVOTools repo commit b3d4494892c6d85fa2a35d548271fb02ea3e191b	Upstream stateful REPL/retry/verifier semantics.
S6	PiVOTools agentic/system blobs dab87e08… / 265131a5…	Exact embedded prompt provenance.
S7	PIV-Q27-P0 v1.7 CROSS-AUDITED + SMOKE bundle	Proves XHIGH PIV activation failure; also contains scheduler hardening.
S8	Reasoning activation v1.9d / GPU probe 20260915T124541Z	LOW and BUDGET3072 tool activation; MEDIUM weak.
S9	Qwen3.8 official chat template / quickstart	Effort values xhigh/medium/low and documented thinking-budget

		continuation.
S10	vLLM 0.19 documentation/source	Serving kwargs, tokenized prompts, cache/topology/runtime behavior.
S11	TP-Q27 v1.2.4 SMOKE evidence ZIP SHA c34fd373…e2aa	Latest real 4×L4 serving measurements and harness failure evidence.
S12	P1-SPARSE evidence + executed notebook	Secondary matched-replication-needed branch; not main solver line.

Important version/hashing notes

- Q1 frozen Proxy-48 ordered SHA256: 8f432c60a35091dcda2b890574b1f4d3c03aaf48a5b3b7f96b4e88c35d359134.
- PIV v1.6 external notebook SHA: 7b7d94c31534476ef50cb84b36f0e250b82c70dfc4504e909c1c27c6865195c0; superseded by v1.7 after multi-call bug fix.
- PIV v1.7 CROSS-AUDITED SHA: 61d4e2095111c08ab16c361fadf1a4ad054363eb35c52ee10b450c8221ffdc6a.
- External ACT v1.8 SHA: 7245e7df5c684e3d59d8af390b9748c391e6b1ad46e420001f9bf5d113c2b586; later iterations fixed continuation/recommendation/gating issues.
- Throughput v1.2.4 latest smoke evidence SHA: c34fd3736dd9d2565ef681c6fd1936bb7eb1f99b3a95d222eda419049cd3e2aa.

13. New-chat kickoff prompt

নতুন chat-এর প্রথম message হিসেবে নিচের condensed brief paste করা যায়:

```
We are continuing an ARC-AGI-2 Qwen3.8-27B-FP8 project on Kaggle 4xL4. Treat the attached handoff PDF as
canonical history.

Current verified state:
- Q1 Direct OFF/XHIGH: 0 exact; XHIGH 48/48 hit 12,288 cap.
- PIV v1.7 XHIGH SMOKE: 8/8 hit cap, 0 tool calls -> do NOT run that Proxy48.
- Reasoning activation probe: LOW 0.875 tool engagement, MEDIUM 0.125, BUDGET3072 0.875; LOW+BUDGET
approved for further solving tests.
- Latest throughput SMOKE v1.2.4: CUDA graph/noAR long@8 129.17 tok/s vs frozen 48.30 (~2.67x); custom
AR no gain.
- Prefix64 and TP2 OOMs are inconclusive because concurrency stress was too aggressive.
- Current 0.50 greedy-prefix numerics gate is invalid: unchanged C0 repeat scored 0.1875.
- Current broad PROBE_GPU must NOT be run.

Immediate priority:
1) C1 graph/noAR real solver smoke (only engine change) with fresh all-train exact verification.
2) New focused fail-fast throughput follow-up: prefix at mns16, cache A/B before stress, TP2 n=2/4/8
with lower batched-token budgets, then TP2x2 only from safe points.
3) Compare LOW vs BUDGET3072 on verified program quality, not engagement alone.
4) Once non-zero all-train-exact appears, advance to Proxy/Public-48, multi-sample pass@K, then
Public-120.

Do not resurrect Qwen 'high', PIV-v1.7 XHIGH Proxy48, the old broad throughput probe, raw-cell
similarity as correctness, or the broken greedy-prefix numerics gate. Source-check and validate every
notebook after changes.
```

Final handoff verdict

Where the project really stands: We are no longer debugging basic dataset/scoring/parser infrastructure. The key model behavior and serving bottlenecks are now identified. The next decisive milestone is not another harness metric: it is the first reproducible non-zero all-train-exact program and official exact ARC solve under the faster C1 serving path and a working reasoning-control arm. Once that appears, multi-sample search becomes worth scaling.

End of handoff. Preserve this PDF with the corresponding executed notebooks/bundles; future conclusions should always identify whether they are source-verified, harness-verified, real-GPU-measured, or only inferred.